

Suggested solution for the open question

ECO360

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Here I provide a full and detailed solution for the open question. If you attempted the bonus question, I don't expect to see solutions as detailed as this one, but you should at least provide a sound argument or some intuition of any claim you make.

PART II: OPEN QUESTION (BONUS)

Let p_t be the price of an asset with variable dividends d_t across time. Dividends d_t are obtained at the end of period t . Let r be the interest rate of a risk-free asset that is constant across time.

1. Suppose that the buyer buys the asset at price p_t . What are the returns of the asset at the end of period $t + 1$? What is the *rate of return* of this asset at the end of period $t + 1$? (2 points)

Solution: If the buyer buys the asset at price p_t , then this investment strategy yields at the end of period $t + 1$:

$$d_t + (p_{t+1} - p_t),$$

since the buyer gets the dividends and can sell the asset at a price p_{t+1} . Hence, the rate of return is equal to

$$R_t = \frac{d_t + (p_{t+1} - p_t)}{p_t}.$$

□

2. Suppose that instead of spending p_t to buy the dividend bearing asset, the buyer uses this money to invest in the risk-free asset. What are the returns on this risk-free asset at the end of period $t + 1$. What is the rate of return on the risk-free asset at the end of period $t + 1$? (2 points)

Solution: If instead the buyer uses an amount p_t to invest in the risk-free asset, then she gets a return at the end of period $t + 1$ of:

$$rp_t,$$

thus the rate of return is just r .

□

3. Assuming that there are *no arbitrage opportunities*, what condition must hold about these two investment strategies? (2 points)

Solution: If there are no arbitrage opportunities, then the returns or the rate of returns on both assets must be the same, thus:

$$d_t + (p_{t+1} - p_t) = p_t r, \quad \text{or} \quad R_t = r. \quad (1)$$

□

4. Suppose that there is no uncertainty, so the buyer has perfect information about the sequence of prices $\{p_t\}$ and dividends $\{d_t\}$ that will prevail in the future. Using the no-arbitrage condition derived in the previous question, derive an equation that relates p_t , p_{t+1} and d_t . (2 points)

Solution: Using the no-arbitrage condition (1) we get:

$$p_t = \frac{1}{1+r}d_t + \frac{1}{1+r}p_{t+1}. \quad (2)$$

□

5. Assume that there are no speculative bubbles. Solve the equation you found in part (4). You should get an expression that relates the current price p_t with the discounted value of the dividends. (2 points)

Solution: Iterate forward one period equation (2) to get

$$p_{t+1} = \frac{1}{1+r}d_{t+1} + \frac{1}{1+r}p_{t+2}.$$

Plug the previous equation into (2) to get:

$$p_t = \frac{1}{1+r}d_t + \frac{1}{1+r} \left[\frac{1}{1+r}d_{t+1} + \frac{1}{1+r}p_{t+2} \right] = \frac{1}{1+r}d_t + \left(\frac{1}{1+r} \right)^2 d_{t+1} + \left(\frac{1}{1+r} \right)^2 p_{t+2},$$

continuing iterating forward until period T and plugging the result into the equation yields:

$$p_t = \frac{1}{1+r} \sum_{k=0}^{T-1} \left(\frac{1}{1+r} \right)^k d_{t+k} + \left(\frac{1}{1+r} \right)^T p_{t+T},$$

which taking limits when $T \rightarrow \infty$ yields

$$p_t = \frac{1}{1+r} \sum_{k=0}^{\infty} \left(\frac{1}{1+r} \right)^k d_{t+k} + \lim_{T \rightarrow \infty} \left(\frac{1}{1+r} \right)^T p_{t+T},$$

since there are no speculative bubbles we know that $\lim_{T \rightarrow \infty} \left(\frac{1}{1+r} \right)^T p_{t+T} = 0$, thus

$$p_t = \frac{1}{1+r} \sum_{k=0}^{\infty} \left(\frac{1}{1+r} \right)^k d_{t+k}.$$

Thus, what we get is that in the absence of speculative bubbles, the price of the dividend bearing asset at time t is equal to the present value of its dividend stream. □